

**SIMATS**  
ENGINEERING

**SIMATS Online Programming Lab**  
JAVA PROGRAMMING

Bheeshma L  
192511332RC

QUESTIONS

Q1  
Matrix multiplication  
10 marks

Q2  
Largest number  
10 marks

Q3  
Smallest number  
10 marks

Q4  
With largest number  
10 marks

Q5  
Palindrome number/string  
10 marks

Q6  
Removing duplicates in array  
10 marks

6/6 answered

Q1 Matrix multiplication 10 marks

Matrix multiplication

Your Code

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         if (!sc.hasNextInt()) return;  
8  
9         int r1 = sc.nextInt();  
10        int c1 = sc.nextInt();  
11        int[][] a = new int[r1][c1];  
12        for (int i = 0; i < r1; i++) {  
13            for (int j = 0; j < c1; j++) {  
14                a[i][j] = sc.nextInt();  
15            }  
16        }  
17  
18        int r2 = sc.nextInt();  
19        int c2 = sc.nextInt();  
20        int[][] b = new int[r2][c2];  
21        for (int i = 0; i < r2; i++) {  
22            for (int j = 0; j < c2; j++) {  
23                b[i][j] = sc.nextInt();  
24            }  
25        }  
26  
27        if (c1 != r2) {  
28            System.out.println("Multiplication not possible.");  
29            return;  
30        }  
31  
32        int[][] c = new int[r1][c2];  
33        for (int i = 0; i < r1; i++) {  
34            for (int j = 0; j < c2; j++) {  
35                for (int k = 0; k < c1; k++) {  
36                    c[i][j] += a[i][k] * b[k][j];  
37                }  
38            }  
39        }  
40  
41        for (int i = 0; i < r1; i++) {  
42            for (int j = 0; j < c2; j++) {  
43                System.out.print(c[i][j] + " ");  
44            }  
45            System.out.println();  
46        }  
47    }  
}
```

Input

2 3  
1 2 3  
4 5 6  
3 2  
...

Output

21 19  
85 55

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PRACTICE PROGRAMS

**QUESTIONS**  

**Q1**  
Matrix multiplication  
10 marks

**Q2**  
. Largest number  
10 marks

**Q3**  
Smallest number  
10 marks

**Q4**  
. Nth largest number  
10 marks

**Q5**  
Palindrome number/string  
10 marks

**Q6**  
Removing duplicates in array  
10 marks

6/6 answered

**Q2 . Largest number** 10 marks

. Largest number

**Your Code**

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (!sc.hasNextInt()) return;
8         int n = sc.nextInt();
9
10
11         int[] arr = new int[n];
12         for (int i = 0; i < n; i++) {
13             arr[i] = sc.nextInt();
14         }
15
16         int max = arr[0];
17         for (int i = 1; i < n; i++) {
18             if (arr[i] > max) {
19                 max = arr[i];
20             }
21         }
22         System.out.println(max);
23     }
24 }
25 }
```

**Input**  
5  
17 98 56 66 23

**Output**  
98

▶ Run

☁ Save

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PRACTICE PROGRAMS

**QUESTIONS**  

**Q1**  
Matrix multiplication  
10 marks

**Q2**  
. Largest number  
10 marks

**Q3**  
Smallest number  
10 marks

**Q4**  
. Nth largest number  
10 marks

**Q5**  
Palindrome number/string  
10 marks

**Q6**  
Removing duplicates in array  
10 marks

6/6 answered

**Q3 Smallest number** 10 marks

Smallest number

**Your Code**

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (!sc.hasNextInt()) return;
8         int n = sc.nextInt();
9
10
11         int[] arr = new int[n];
12         for (int i = 0; i < n; i++) {
13             arr[i] = sc.nextInt();
14         }
15
16         int min = arr[0];
17         for (int i = 1; i < n; i++) {
18             if (arr[i] < min) {
19                 min = arr[i];
20             }
21         }
22         System.out.println(min);
23     }
24 }
25 }
```

**Input**  
5  
88 56 3 31 11

**Output**

▶ Run

☁ Save

PRACTICE PROGRAMS

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QUESTIONS

Q1  
Matrix multiplication  
10 marks

Q2  
. Largest number  
10 marks

Q3  
Smallest number  
10 marks

Q4  
. Nth largest number  
10 marks

Q5  
Palindrome number/string  
10 marks

Q6  
Removing duplicates in array  
10 marks

6/6 answered

Q4 . Nth largest number 10 marks

. Nth largest number

Your Code

```
1 import java.util.Scanner;
2 import java.util.Arrays;
3
4 public class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         if (!sc.hasNextInt()) return;
9
10        int size = sc.nextInt();
11        int[] arr = new int[size];
12
13        for (int i = 0; i < size; i++) {
14            arr[i] = sc.nextInt();
15        }
16
17        int n = sc.nextInt();
18
19        Arrays.sort(arr);
20
21        if (n > 0 && n <= size) {
22            System.out.println(arr[size - n]);
23        }
24    }
25 }
```

Run Save

Input

5  
11 22 33 44 55  
2

Output

PRACTICE PROGRAMS

← Back

QUESTIONS

Q1  
Matrix multiplication  
10 marks

Q2  
. Largest number  
10 marks

Q3  
Smallest number  
10 marks

Q4  
. Nth largest number  
10 marks

Q5  
Palindrome number/string  
10 marks

Q6  
Removing duplicates in array  
10 marks

6/6 answered

Q5 Palindrome number/string 10 marks

Palindrome number/string

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (!sc.hasNext()) return;
8
9         String input = sc.next();
10        String reversed = new StringBuilder(input).reverse().toString();
11
12        if (input.equals(reversed)) {
13            System.out.println("Palindrome");
14        } else {
15            System.out.println("Not Palindrome");
16        }
17    }
18 }
```

Run Save

Input

malayalam

Output

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192511332R

**PRACTICE PROGRAMS**← Back

**QUESTIONS**

**Q1**  
Matrix multiplication  
10 marks

**Q2**  
. Largest number  
10 marks

**Q3**  
Smallest number  
10 marks

**Q4**  
. Nth largest number  
10 marks

**Q5**  
Palindrome number/string  
10 marks

**Q6**  
Removing duplicates in array  
10 marks

6/6 answered

**Q6 Removing duplicates in array**10 marks

Removing duplicates in array

**Your Code**

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (!sc.hasNextInt()) return;
8
9         int n = sc.nextInt();
10        int[] arr = new int[n];
11
12        for (int i = 0; i < n; i++) {
13            arr[i] = sc.nextInt();
14        }
15
16        int[] temp = new int[n];
17        int j = 0;
18
19        for (int i = 0; i < n; i++) {
20            boolean isDuplicate = false;
21            for (int k = 0; k < j; k++) {
22                if (arr[i] == temp[k]) {
23                    isDuplicate = true;
24                    break;
25                }
26            }
27            if (!isDuplicate) {
28                temp[j++] = arr[i];
29            }
30        }
31
32        for (int i = 0; i < j; i++) {
33            System.out.print(temp[i] + (i < j - 1 ? " " : ""));
34        }
35    }
36 }
```

**Input**

6  
10 10 10 20 30 40

**Output**

10 20 30 40